

One-Sided, Infinite, and End-Behavior Limits

AP Calculus AB / BC

One-sided limits, infinite limits at a vertical asymptote, end behavior at infinity, and continuity of a piecewise function. Decide which of the three situations you are in before computing:

- **One-sided.** The superscript $+$ or $-$ tells you which branch of the function to use and which sign a small factor carries.
- **Infinite.** A nonzero numerator over a denominator going to zero is $\pm\infty$; get the sign from the sign of the denominator on that side. Write ∞ or $-\infty$, or “does not exist” with a reason — never a number.
- **End behavior.** As $x \rightarrow \pm\infty$, compare degrees, or divide numerator and denominator by the highest power of x in the denominator.

Show the reasoning line that decides the answer, not just the answer.

1. Evaluate: $\lim_{x \rightarrow 2^+} \frac{|x - 2|}{x - 2}$

Answer: _____

2. Evaluate: $\lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 1}{x^2 - 4}$

Answer: _____

3. Evaluate: $\lim_{x \rightarrow 5^-} \frac{1}{x-5}$

State the sign of $x - 5$ for values just below 5 as part of your reasoning.

Answer: _____

4. Evaluate: $\lim_{x \rightarrow -\infty} \frac{4x^3 - x}{2x^3 + 7}$

Answer: _____

5. Let

$$f(x) = \begin{cases} x^2 + 1, & x < 2 \\ 5x - 3, & x \geq 2 \end{cases}$$

(a) Find $\lim_{x \rightarrow 2^-} f(x)$. (b) Find $\lim_{x \rightarrow 2^+} f(x)$. (c) Does $\lim_{x \rightarrow 2} f(x)$ exist? Explain in one sentence.

(a) _____ (b) _____

6. Evaluate: $\lim_{x \rightarrow -4^+} \frac{2x}{x+4}$

Answer: _____

7. Evaluate: $\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + 4}}{2x + 1}$

Divide by the highest power of x in the denominator, and be explicit about what $\sqrt{x^2}$ equals when x is large and positive.

Answer: _____

8. Evaluate: $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x^2} \right)$

Combine over a common denominator first — the two pieces separately are both infinite, and their difference is not settled until they are combined.

Answer: _____

9. Let

$$g(x) = \begin{cases} x^2 + k, & x < 3 \\ 4x - 1, & x \geq 3 \end{cases}$$

Find the value of the constant k that makes g continuous at $x = 3$. Write the one-sided limits first, set them equal, and solve.

Answer: _____

10. Synthesis challenge. Let $h(x) = \frac{2x^2 - 3x}{x^2 - x - 6}$.

(a) Find $\lim_{x \rightarrow \infty} h(x)$. (b) Find $\lim_{x \rightarrow 3^-} h(x)$. (c) Find $\lim_{x \rightarrow 3^+} h(x)$. (d) Using your answers, describe every vertical and horizontal asymptote of the graph of h in words.

(a) _____ (b) _____ (c) _____